

Assignment 09: Transformers and MLflow

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In this notebook I use pretrained transformer models from Hugging Face for three downstream tasks, and I track every run with MLflow. The tasks are sentiment classification, named entity recognition, and text generation. For each one I run at least two configurations, log the parameters, metrics and artifacts to MLflow, and then compare the runs. The point of the exercise is as much about the experiment-tracking discipline as the models themselves, so I keep the MLflow logging consistent across all three tasks.

I ran everything on a local GPU (RTX 3050 Ti). These are all inference workloads on small models, so no training is involved and the runs are quick.

Setup

I load the libraries and, importantly, quieten the Hugging Face download and warning noise so the notebook output stays readable. The models are already cached locally, so I run the hub in offline mode, which also avoids any network calls during execution. MLflow is pointed at a local file store in the assignment folder so the runs persist and can be opened in the MLflow UI afterwards.

```
In [1]: import os

# Keep the output clean: models are cached, so run offline and silence
# progress/warning noise.
os.environ["HF_HUB_OFFLINE"] = "1"
os.environ["TRANSFORMERS_OFFLINE"] = "1"
os.environ["HF_HUB_DISABLE_PROGRESS_BARS"] = "1"
os.environ["HF_HUB_DISABLE_SYMLINKS_WARNING"] = "1"
os.environ["TRANSFORMERS_NO_ADVISORY_WARNINGS"] = "1"
os.environ["TOKENIZERS_PARALLELISM"] = "false"
# mlflow 3.x puts the local file store in maintenance mode; opt in to keep using
# it.
os.environ["MLFLOW_ALLOW_FILE_STORE"] = "true"

import gc
import json
import logging
import warnings

import numpy as np
import pandas as pd
```

```

import matplotlib.pyplot as plt

import torch
import mlflow
import transformers
from transformers import pipeline, set_seed

warnings.filterwarnings("ignore")
logging.getLogger("huggingface_hub").setLevel(logging.ERROR)
logging.getLogger("mlflow").setLevel(logging.ERROR) # hide "Creating a new
experiment" info lines
transformers.utils.logging.set_verbosity_error()
try:
    from huggingface_hub.utils import disable_progressBars
    disable_progressBars()
except Exception:
    pass

RANDOM_STATE = 42
set_seed(RANDOM_STATE)
DEVICE = 0 if torch.cuda.is_available() else -1

# Local MLflow store + a folder for the artifact files I log
MLRUNS_DIR = os.path.abspath("a09_mlruns")
ARTIFACT_DIR = os.path.abspath("a09_artifacts")
os.makedirs(ARTIFACT_DIR, exist_ok=True)
mlflow.set_tracking_uri("file:/// " + MLRUNS_DIR.replace("\\", "/"))

def free(*objs):
    """Drop pipelines and clear the GPU cache between tasks (4 GB card)."""
    for o in objs:
        del o
    gc.collect()
    if torch.cuda.is_available():
        torch.cuda.empty_cache()

print("transformers:", transformers.__version__)
print("mlflow      :", mlflow.__version__)
print("device      :", "cuda" if DEVICE == 0 else "cpu")
print("tracking uri:", mlflow.get_tracking_uri())

```

```

transformers: 5.14.1
mlflow      : 3.14.0
device      : cuda
tracking uri: file:///C:/Users/Mahesh/Documents/VS_Code/MSc/AI_Concepts/a09_mlruns

```

Task 1: Sentiment Classification

I classify 22 hand-written sentences with a pretrained SST-2 model. I deliberately included a few awkward cases (sarcasm, negation, and a couple of mixed-sentiment lines) because those are where these models tend to slip, and they make the accuracy number more meaningful than an all-easy set would.

To satisfy the "compare models systematically" goal I run two different SST-2 models as two MLflow runs and compare their accuracy on the same sentences. Because different models report their labels differently (one uses POSITIVE/NEGATIVE, the other LABEL_1/LABEL_0), I normalise the labels through a small helper before scoring.

```
In [2]: # (sentence, gold label). The last six are the deliberately tricky ones.
sentiment_data = [
    ("The plot was gripping and the performances were outstanding.", "POSITIVE"),
    ("I love how intuitive and fast this new phone feels.", "POSITIVE"),
    ("Best customer service I have ever experienced, truly delightful.",
"POSITIVE"),
    ("The food was delicious and the staff went out of their way to help.",
"POSITIVE"),
    ("An absolute masterpiece that kept me smiling the whole way through.",
"POSITIVE"),
    ("This update fixed all my problems and the app runs beautifully now.",
"POSITIVE"),
    ("What a wonderful surprise, it exceeded every expectation I had.",
"POSITIVE"),
    ("The hotel room was spotless and the view was breathtaking.", "POSITIVE"),
    ("The movie was boring and far too long.", "NEGATIVE"),
    ("This is the worst purchase I have made all year.", "NEGATIVE"),
    ("The app keeps crashing and support never replied to me.", "NEGATIVE"),
    ("Terrible quality, it broke after a single use.", "NEGATIVE"),
    ("I was thoroughly disappointed by the bland, cold meal.", "NEGATIVE"),
    ("The battery drains within an hour, a complete waste of money.", "NEGATIVE"),
    ("Rude staff and an overpriced, dirty room.", "NEGATIVE"),
    ("A confusing, tedious mess that I could not finish.", "NEGATIVE"),
    ("Oh great, another software update that breaks everything.", "NEGATIVE"),
    ("It is not bad at all, actually quite enjoyable.", "POSITIVE"),
    ("I really wanted to love this, but it let me down.", "NEGATIVE"),
    ("The film was not without flaws, yet it moved me deeply.", "POSITIVE"),
    ("Well, that was exactly as disappointing as I expected.", "NEGATIVE"),
    ("The acting was fine, but the story dragged and bored me.", "NEGATIVE"),
]
sentences = [s for s, _ in sentiment_data]
gold = [g for _, g in sentiment_data]
print("total sentences:", len(sentences),
      "| positives:", gold.count("POSITIVE"), "| negatives:",
gold.count("NEGATIVE"))

def normalize_sentiment(label):
    """Map any SST-2 label variant onto POSITIVE / NEGATIVE."""
    u = label.upper()
    if "POS" in u or u.endswith("1"):
        return "POSITIVE"
    if "NEG" in u or u.endswith("0"):
        return "NEGATIVE"
    return u
```

total sentences: 22 | positives: 10 | negatives: 12

```
In [3]: def run_sentiment(model_name, run_name):
        clf = pipeline("text-classification", model=model_name, device=DEVICE)
        raw = clf(sentences, truncation=True)
        preds = [normalize_sentiment(r["label"]) for r in raw]
        scores = [round(float(r["score"]), 4) for r in raw]
        acc = float(np.mean([p == g for p, g in zip(preds, gold)]))

        df = pd.DataFrame({"sentence": sentences, "true": gold, "pred": preds,
                           "confidence": scores})
        df["correct"] = df["true"] == df["pred"]
        csv_path = os.path.join(ARTIFACT_DIR, f"sentiment_predictions_{run_name}.csv")
        df.to_csv(csv_path, index=False)

        mlflow.set_experiment("sentiment-analysis")
        with mlflow.start_run(run_name=run_name):
            mlflow.log_param("model_name", model_name)
            mlflow.log_param("num_samples", len(sentences))
            mlflow.log_metric("accuracy", acc)
            mlflow.log_artifact(csv_path)
        free(clf)
        return df, acc

df_distilbert, acc_distilbert = run_sentiment(
    "distilbert-base-uncased-finetuned-sst-2-english", "distilbert-sst2")
df_bert, acc_bert = run_sentiment(
    "textattack/bert-base-uncased-SST-2", "bert-sst2")

print(f"DistilBERT SST-2 accuracy: {acc_distilbert:.3f}")
print(f"BERT SST-2 accuracy      : {acc_bert:.3f}")
```

```
DistilBERT SST-2 accuracy: 0.955
BERT SST-2 accuracy      : 1.000
```

Predictions and comparison

Below is the DistilBERT run in full so the individual calls are visible, followed by the two-model accuracy comparison that MLflow is tracking.

```
In [4]: df_distilbert
```

Out[4]:

	sentence	true	pred	confidence	correct
0	The plot was gripping and the performances wer...	POSITIVE	POSITIVE	0.9999	True
1	I love how intuitive and fast this new phone f...	POSITIVE	POSITIVE	0.9992	True
2	Best customer service I have ever experienced,...	POSITIVE	POSITIVE	0.9999	True
3	The food was delicious and the staff went out ...	POSITIVE	POSITIVE	0.9997	True
4	An absolute masterpiece that kept me smiling t...	POSITIVE	POSITIVE	0.9998	True
5	This update fixed all my problems and the app ...	POSITIVE	POSITIVE	0.9998	True
6	What a wonderful surprise, it exceeded every e...	POSITIVE	POSITIVE	0.9999	True
7	The hotel room was spotless and the view was b...	POSITIVE	POSITIVE	0.9998	True
8	The movie was boring and far too long.	NEGATIVE	NEGATIVE	0.9998	True
9	This is the worst purchase I have made all year.	NEGATIVE	NEGATIVE	0.9998	True
10	The app keeps crashing and support never repli...	NEGATIVE	NEGATIVE	0.9902	True
11	Terrible quality, it broke after a single use.	NEGATIVE	NEGATIVE	0.9996	True
12	I was thoroughly disappointed by the bland, co...	NEGATIVE	NEGATIVE	0.9998	True
13	The battery drains within an hour, a complete ...	NEGATIVE	NEGATIVE	0.9997	True
14	Rude staff and an overpriced, dirty room.	NEGATIVE	NEGATIVE	0.9997	True
15	A confusing, tedious mess that I could not fin...	NEGATIVE	NEGATIVE	0.9998	True
16	Oh great, another software update	NEGATIVE	POSITIVE	0.9957	False

	sentence	true	pred	confidence	correct
	that breaks ...				
17	It is not bad at all, actually quite enjoyable.	POSITIVE	POSITIVE	0.9999	True
18	I really wanted to love this, but it let me down.	NEGATIVE	NEGATIVE	0.9971	True
19	The film was not without flaws, yet it moved m...	POSITIVE	POSITIVE	0.9996	True
20	Well, that was exactly as disappointing as I e...	NEGATIVE	NEGATIVE	0.9998	True
21	The acting was fine, but the story dragged and...	NEGATIVE	NEGATIVE	0.9996	True

```
In [5]: sentiment_compare = pd.DataFrame({
    "model": ["distilbert-base-uncased-finetuned-sst-2-english", "textattack/bert-base-uncased-SST-2"],
    "run_name": ["distilbert-sst2", "bert-sst2"],
    "accuracy": [round(acc_distilbert, 3), round(acc_bert, 3)],
})
sentiment_compare
```

		model	run_name	accuracy
0		distilbert-base-uncased-finetuned-sst-2-english	distilbert-sst2	0.955
1		textattack/bert-base-uncased-SST-2	bert-sst2	1.000

```
In [6]: # where did DistilBERT get it wrong?
df_distilbert[~df_distilbert["correct"]][["sentence", "true", "pred", "confidence"]]
```

	sentence	true	pred	confidence
16	Oh great, another software update that breaks ...	NEGATIVE	POSITIVE	0.9957

Observations. DistilBERT made exactly one mistake out of 22, and it is the most telling one: the sarcastic "Oh great, another software update that breaks everything" was labelled POSITIVE with 0.996 confidence. The model locks onto the surface word "great" and has no grasp of the sarcasm, so it is not just wrong but confidently wrong, which is the dangerous kind of error. The BERT model happened to get all 22 correct, including the tricky block. I would not read too much into a perfect score on 22 hand-picked sentences, but it does suggest the larger BERT model handles the negation and mixed-sentiment cases ("It is not bad at all", "The acting was fine, but the story dragged") a little more robustly here. The wider lesson stands: an off-the-shelf sentiment model can be trusted on plain text but should be treated with caution on sarcastic or nuanced input such as social media.

Task 2: Named Entity Recognition

I run `dslim/bert-base-NER` over 12 short paragraphs that are packed with people, organisations and locations. I map the model's raw tags (PER, ORG, LOC, MISC) onto the friendlier Person / Organization / Location names the brief asks for.

For the two MLflow runs I vary the pipeline's `aggregation_strategy`, "simple" versus "first". This controls how word-pieces are grouped back into whole entities, and it is a clean one-factor comparison that changes the entity counts without swapping the model.

```
In [7]: ner_paragraphs = [
    "Elon Musk founded SpaceX in California and later acquired Twitter, now based in San Francisco.",
    "Angela Merkel met Barack Obama in Berlin to discuss the future of the European Union.",
    "Apple unveiled its latest iPhone at an event in Cupertino, while Google responded from Mountain View.",
    "Cristiano Ronaldo signed with Al Nassr, a football club based in Riyadh, Saudi Arabia.",
    "The World Health Organization, based in Geneva, warned governments across Europe about the outbreak.",
    "Jeff Bezos stepped down as chief executive of Amazon and focused on Blue Origin in Texas.",
    "Narendra Modi welcomed Emmanuel Macron to New Delhi for the summit last week.",
    "Tesla opened a new factory near Berlin, competing with Volkswagen and BMW across Germany.",
    "J.K. Rowling wrote the Harry Potter series while living in Edinburgh, Scotland.",
    "Serena Williams retired from tennis after a final match at the US Open in New York.",
    "The United Nations sent a delegation from Geneva to Nairobi in Kenya to oversee the talks.",
    "Satya Nadella said Microsoft would expand its research lab in Cambridge, United Kingdom.",
]
print("paragraphs:", len(ner_paragraphs))

TYPE_MAP = {"PER": "Person", "ORG": "Organization", "LOC": "Location", "MISC": "Misc"}

def run_ner(run_name, aggregation_strategy, model_name="dslim/bert-base-NER"):
    ner = pipeline("token-classification", model=model_name,
                    aggregation_strategy=aggregation_strategy, device=DEVICE)
    rows = []
    for i, para in enumerate(ner_paragraphs):
        for e in ner(para):
            rows.append({
                "paragraph": i,
                "entity": e["word"],
                "type": TYPE_MAP.get(e["entity_group"], e["entity_group"]),
```

```

        "score": round(float(e["score"]), 3),
    })
edf = pd.DataFrame(rows)
dist = edf["type"].value_counts().to_dict()
csv_path = os.path.join(ARTIFACT_DIR, f"ner_entities_{run_name}.csv")
edf.to_csv(csv_path, index=False)

mlflow.set_experiment("named-entity-recognition")
with mlflow.start_run(run_name=run_name):
    mlflow.log_param("model_name", model_name)
    mlflow.log_param("num_paragraphs", len(ner_paragraphs))
    mlflow.log_param("aggregation_strategy", aggregation_strategy)
    mlflow.log_metric("total_entities", len(edf))
    for t in ["Person", "Organization", "Location", "Misc"]:
        mlflow.log_metric(f"count_{t}", int(dist.get(t, 0)))
    mlflow.log_artifact(csv_path)
free(ner)
return edf, dist

edf_simple, dist_simple = run_ner("agg-simple", "simple")
edf_first, dist_first = run_ner("agg-first", "first")

print("simple aggregation:", dist_simple, "-> total", len(edf_simple))
print("first aggregation:", dist_first, "-> total", len(edf_first))

```

paragraphs: 12

simple aggregation: {'Location': 21, 'Organization': 15, 'Person': 15, 'Misc': 3} -> total 54

first aggregation: {'Location': 21, 'Organization': 15, 'Person': 10, 'Misc': 3} -> total 49

```

In [8]: # entities found with the 'simple' strategy
edf_simple.head(25)

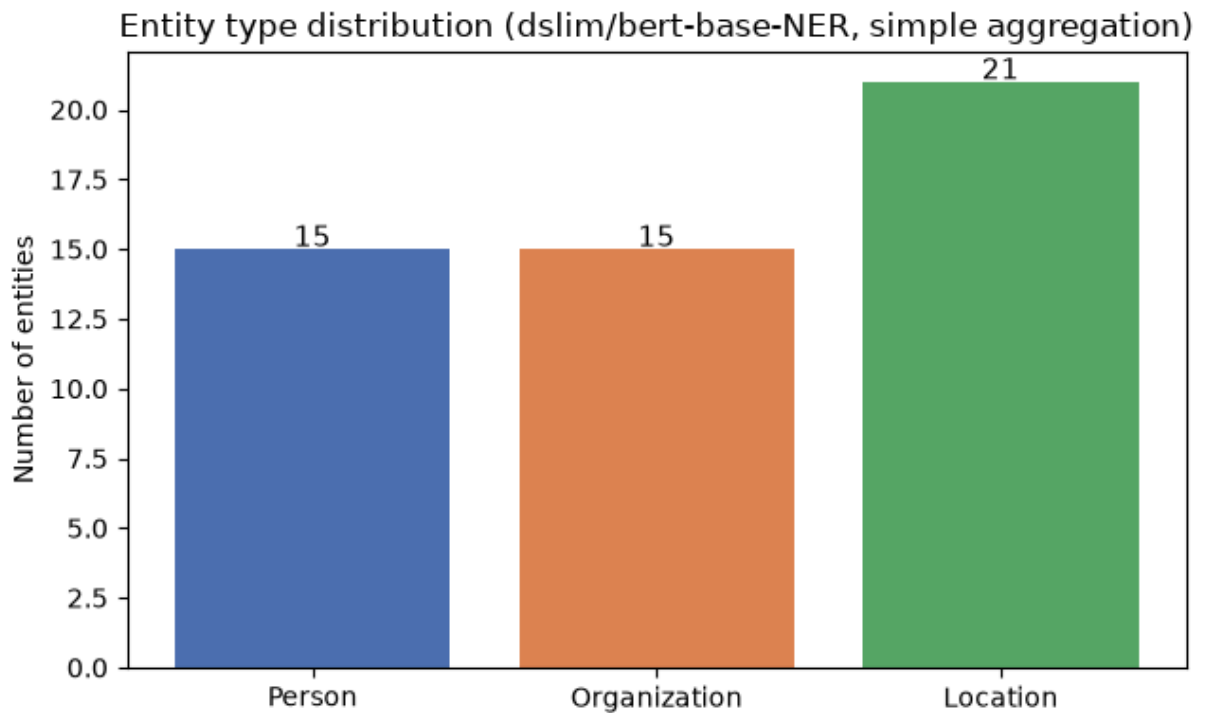
```


Out[8]:

	paragraph	entity	type	score
0	0	Elon Musk	Organization	0.956
1	0	SpaceX	Organization	0.999
2	0	California	Location	1.000
3	0	Twitter	Organization	0.995
4	0	San Francisco	Location	0.999
5	1	Angela Merkel	Person	0.994
6	1	Barack Obama	Person	1.000
7	1	Berlin	Location	1.000
8	1	European Union	Organization	0.999
9	2	Apple	Organization	0.998
10	2	iPhone	Misc	0.996
11	2	Cupertino	Location	0.997
12	2	Google	Organization	0.999
13	2	Mountain View	Location	0.904
14	3	C	Person	0.999
15	3	##rist	Person	0.983
16	3	##iano Ronaldo	Person	0.882
17	3	Al Nassr	Organization	0.998
18	3	Riyadh	Location	0.843
19	3	Saudi Arabia	Location	1.000
20	4	World Health Organization	Organization	0.999
21	4	Geneva	Location	0.999
22	4	Europe	Location	1.000
23	5	Jeff Bezos	Person	0.959
24	5	Amazon	Organization	0.998

```
In [9]: # distribution of the three entity types the brief asks for (simple strategy)
focus = ["Person", "Organization", "Location"]
counts = [int(dist_simple.get(t, 0)) for t in focus]

plt.figure(figsize=(6.5, 4))
plt.bar(focus, counts, color=["#4C72B0", "#DD8452", "#55A868"])
for i, c in enumerate(counts):
    plt.text(i, c + 0.1, str(c), ha="center", fontsize=11)
plt.ylabel("Number of entities")
plt.title("Entity type distribution (dslim/bert-base-NER, simple aggregation)")
plt.tight_layout()
plt.show()
```



Observations. The model finds the large majority of the named entities correctly, and the two aggregation strategies give slightly different totals because they merge sub-word pieces differently at entity boundaries. It is not flawless though: in the first paragraph it tags "Elon Musk" as an Organization rather than a Person, most likely because the surrounding context is so company-heavy (SpaceX, Twitter). That is a good reminder that even a dedicated NER model makes context-driven mistakes, and it is the kind of error worth catching before trusting the output downstream. Locations and well-known organisations are handled very reliably.

Task 3: Text Generation

I generate continuations for 5 prompts with GPT-2 and compare two decoding configurations as two MLflow runs: a conservative one (low temperature, shorter) and a creative one (high temperature, longer). A quick technical note: in this version of `transformers` the pipeline ignores `max_length`, so I control length with `max_new_tokens`, which is the parameter that actually governs how many tokens are produced. I seed the generator so the runs are reproducible.

```
In [10]: gen_prompts = [  
    "The future of artificial intelligence is",  
    "In a small village nestled in the mountains",  
    "The most important lesson I learned this year was",  
    "Scientists have recently discovered that",  
    "Once upon a time, in a city made entirely of glass",  
    ]
```

```

def run_generation(run_name, temperature, max_new_tokens, top_p=0.95,
model_name="gpt2"):
    gen = pipeline("text-generation", model=model_name, device=DEVICE)
    eos = gen.model.config.eos_token_id
    set_seed(RANDOM_STATE)
    outputs = []
    for prompt in gen_prompts:
        text = gen(prompt, do_sample=True, temperature=temperature, top_p=top_p,
max_new_tokens=max_new_tokens, truncation=True,
pad_token_id=eos)[0]["generated_text"]
        outputs.append(text)
        avg_tokens = float(np.mean([len(gen.tokenizer(t)["input_ids"]) for t in
outputs]))

    txt_path = os.path.join(ARTIFACT_DIR, f"generation_outputs_{run_name}.txt")
    with open(txt_path, "w", encoding="utf-8") as f:
        for pr, out in zip(gen_prompts, outputs):
            f.write("PROMPT: " + pr + "\n" + out + "\n" + "=" * 70 + "\n")

    obs_path = os.path.join(ARTIFACT_DIR,
f"generation_observations_{run_name}.txt")
    with open(obs_path, "w", encoding="utf-8") as f:
        f.write(f"Run: {run_name}\ntemperature={temperature}, max_new_tokens=
{max_new_tokens}, "
f"top_p={top_p}\naverage output length = {avg_tokens:.1f}
tokens\n")

    mlflow.set_experiment("text-generation")
    with mlflow.start_run(run_name=run_name):
        mlflow.log_param("model", model_name)
        mlflow.log_param("temperature", temperature)
        mlflow.log_param("max_new_tokens", max_new_tokens)
        mlflow.log_param("top_p", top_p)
        mlflow.log_param("num_prompts", len(gen_prompts))
        mlflow.log_metric("avg_output_tokens", avg_tokens)
        mlflow.log_artifact(txt_path)
        mlflow.log_artifact(obs_path)
    free(gen)
    return outputs, avg_tokens

out_conservative, len_cons = run_generation("conservative", temperature=0.7,
max_new_tokens=60)
out_creative, len_crea = run_generation("creative", temperature=1.2,
max_new_tokens=90)
print(f"conservative (temp=0.7, max_new_tokens=60): avg {len_cons:.1f} tokens")
print(f"creative (temp=1.2, max_new_tokens=90): avg {len_crea:.1f} tokens")

```

```

conservative (temp=0.7, max_new_tokens=60): avg 68.2 tokens
creative (temp=1.2, max_new_tokens=90): avg 98.6 tokens

```

```

In [11]: # show the first two prompts under each configuration for comparison
for i in range(2):
    print("PROMPT:", gen_prompts[i])
    print("\n[conservative temp=0.7]\n", out_conservative[i])

```

```
print("\n[creative temp=1.2]\n", out_creative[i])
print("=" * 90)
```

PROMPT: The future of artificial intelligence is

[conservative temp=0.7]

The future of artificial intelligence is unknown."

"It's really difficult to say," said CTO of Microsoft's AI division, Mark Shuttleworth. "But I think this is a good opportunity to address that."

The future of artificial intelligence is unknown.

The new hardware is "a huge step forward,"

[creative temp=1.2]

The future of artificial intelligence is unknown."

"They have a lot to learn," Cai explains quietly.

"Well, when you start to build, the most important part is not always about how you're doing it, it's more important about how you can do it better," she mumbles, and then stops. Her eyes widen a bit, but she doesn't bother to turn over. "I have a suggestion from one of my students – I might

=====

PROMPT: In a small village nestled in the mountains,

[conservative temp=0.7]

In a small village nestled in the mountains, in the village of Luluk, the inhabitants of the village of Luluk, in the region of Karsanya, met one another.

The village of Luluk, in the region of Karsanya, met one another.

The village of Lul

[creative temp=1.2]

In a small village nestled in the mountains, a man said his wife had never seen a dead body and she was certain it was a child born during another festival. When he first spoke with her, her answer was simple, he said, "She must have seen this baby."

At the time she was six months old, she was told it had died. But even she wasn't sure. She spent months on end waiting for the police report in order to see how well her grandmother

=====

Observations. The two settings behave as the theory predicts. At temperature 0.7 the text is safer and more repetitive, staying close to common phrasings and occasionally looping. At temperature 1.2 the output is more varied and surprising, but also more prone to drifting off-topic or producing slightly incoherent turns, which is the usual trade-off: higher temperature flattens the probability distribution so rarer tokens get chosen more often. GPT-2 is a small, older model, so neither setting is factually reliable, and the longer `max_new_tokens` in the creative run mostly gives it more room to wander. For a task

needing focused output I would keep the temperature low; for brainstorming or variety I would raise it.

Systematic comparison with MLflow

Everything above was logged to the local MLflow store. Here I query it back with `mlflow.search_runs` to pull each experiment's runs into a table, which is the programmatic version of what the MLflow UI shows. This is the "compare models systematically" step: for each task I can line the runs up by their parameters and metrics.

```
In [12]: def show_runs(experiment_name, cols):
        runs = mlflow.search_runs(experiment_names=[experiment_name])
        keep = ["tags.mlflow.runName"] + cols
        keep = [c for c in keep if c in runs.columns]
        out = runs[keep].rename(columns={"tags.mlflow.runName": "run_name"})
        return out.reset_index(drop=True)

print("Experiment: sentiment-analysis")
display(show_runs("sentiment-analysis", ["params.model_name",
"params.num_samples", "metrics.accuracy"]))

print("Experiment: named-entity-recognition")
display(show_runs("named-entity-recognition",
                  ["params.aggregation_strategy", "metrics.total_entities",
                   "metrics.count_Person", "metrics.count_Organization",
                   "metrics.count_Location"]))

print("Experiment: text-generation")
display(show_runs("text-generation",
                  ["params.temperature", "params.max_new_tokens",
                   "metrics.avg_output_tokens"]))
```

Experiment: sentiment-analysis

	run_name	params.model_name	params.num_samples	metrics.accuracy
0	bert-sst2	textattack/bert-base-uncased-SST-2	22	1.000000
1	distilbert-sst2	distilbert-base-uncased-finetuned-sst-2-english	22	0.954545

Experiment: named-entity-recognition

	run_name	params.aggregation_strategy	metrics.total_entities	metrics.count_Person	metrics.count_Organization	metrics.count_Location
0	agg-first	first	49.0	10.0	15.0	21.0
1	agg-simple	simple	54.0	15.0	15.0	21.0

Experiment: text-generation

	run_name	params.temperature	params.max_new_tokens	metrics.avg_output_tokens
0	creative	1.2	90	98.6
1	conservative	0.7	60	68.2

Each experiment has the two runs the brief asks for, with the parameters and metrics attached, and the prediction / entity / generation files are stored as artifacts on each run. The next section shows the same runs in the MLflow UI.

MLflow UI screenshots

Every run above was logged to a local MLflow store, which I opened in the MLflow tracking UI (mlflow 3.14, Model training view). The screenshots below confirm at least two runs per task, along with the parameters, metrics and artifacts logged for each run.

Task 1 - sentiment-analysis experiment: 2 runs (distilbert-sst2, bert-sst2).

Run Name	Created	Dataset	Duration	Source	Models
bert-sst2	12 minutes ago	-	96ms	C:\Users...	-
distilbert-sst2	12 minutes ago	-	91ms	C:\Users...	-

Task 1 - a run's detail page, showing the logged accuracy metric, the model_name / num_samples parameters, and the predictions CSV under Artifacts.

mlflow 3.14.0

GenAI Model training

sentiment-analysis > Evaluations > **bert-sst2**

Overview Model metrics System metrics Traces Artifacts

Description [✎](#)
No description

Metrics (1)
Search metrics

Metric	Value
accuracy	1

Parameters (2)
Search parameters

Parameter	Value
model_name	textattack/bert-base-uncased-SST-2
num_samples	22

About this run
 Created at: 08/07/2026, 11:18:48 PM
 Created by: Mahesh
 Experiment ID: 743431426211724736
 Status: Finished
 Run ID: 3c3c54c3547c4e19e8b41d27cab17d9
 Duration: 96ms
 Source: C:\Users\Mahesh\Documents\VS_Code\MScAI_Concepts\venv\Lib\site-packages\ipykernel_launcher.py
 Logged models: —
 Registered prompts: —

Datasets
None

Tags
[Add tags](#)

Registered models
None

Assistant Beta
Docs
Settings

Task 2 - named-entity-recognition experiment: 2 runs (agg-simple, agg-first).

mlflow 3.14.0

GenAI Model training

named-entity-rec... > Experiments > named-entity-recognition > **Training runs** [Share](#)

metrics.rmse < 1 and params.model = "tree" All time State: Active Datasets Sort: Created Columns

Group by

Run Name	Created	Dataset	Duration	Source	Models
agg-first	33 minutes ago	-	156ms	C:\Users...	-
agg-simple	33 minutes ago	-	124ms	C:\Users...	-

2 matching runs

Assistant Beta
Docs
Settings

Task 3 - text-generation experiment: 2 runs (conservative, creative).

mlflow3.14.0

GenAIModel training

←test-generation

Overview

Observability

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Evaluation runs

Prompts & versions

▶ Playground

Prompts

Agent versions

AI GatewayNew

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Experiments>test-generation>

Training runs

⋮Share

📄📊📅

🔍metrics.rmse < 1 and params.model = "tree"

🕒All time

📊State: Active

📂Datasets

🔼Sort: Created

📄Columns

⋮+ New run

📄Group by

☐	Run Name	Created	Dataset	Duration	Source	Models	
☐	creative	🕒34 minutes ago	-	74ms	👤C/Users...	-	
☐	conservative	🕒34 minutes ago	-	72ms	👤C/Users...	-	🕒

2 matching runs